THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 224 cm^3	$lbh = 4 \times 8 \times 7$.
2	**(d)** 125 cm^3	5^3.
3	**(d)** 96 cm^2	$6a^2$.
4	**(b)** 84 cm^2	$2h(l + b)$.
5	**(c)** one-third of	$\frac{1}{3}\pi r^2 h$ versus $\pi r^2 h$.
6	**(c)** 2 cm	$\sqrt[3]{8} = 2$.
7	**(b)** 462 cm^3	$\pi r^2 h$ with $r = 7$, $h = 3$.
8	**(b)** 5544 cm^2	$4\pi r^2$ with $r = 21$.
9	**(c)** 162 cm^3	$lbh = 3 \times 9 \times 6$.
10	**(c)** 343 cm^3	7^3.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume $= lbh = 9 \times 6 \times 3 = 162$ cm^3; TSA $= 2(lb + bh + hl) = 2(54 + 18 + 27) = 198$ cm^2 (1); LSA $= 2h(l + b) = 90$ cm^2 (1).	1+1
12	Volume $= 5^3 = 125$ cm^3; TSA $= 6 \times 5^2 = 150$ cm^2 (1); Edge for volume 216: $\sqrt[3]{216} = 6$ cm (1).	1+1
13	Volume $= 8 \times 4 \times 7 = 224$ m^3; Capacity $= 224 \times 1000$ litres (1); $= 224000$ litres (1).	1+1
14	Volume $= lbh = 8 \times 7 \times 2 = 112$ cm^3; TSA $= 2(lb + bh + hl) = 2(56 + 14 + 16) = 172$ cm^2 (1); LSA $= 2h(l + b) = 60$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume $= lbh = 5 \times 15 \times 9 = 675$ cm^3 (1); TSA $= 2(lb + bh + hl) = 2(75 + 135 + 45) = 510$ cm^2 (1); LSA $= 2h(l + b) = 360$ cm^2 (1).	1+1+1
16	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 3.5 \times 11 = 242$ cm^2 (1); TSA $= 2\pi r(r + h) = 319$ cm^2 (1); Volume $= \pi r^2 h = 423.5$ cm^3 (1).	1+1+1
17	Volume $= 17^3 = 4913$ cm^3 (1); TSA $= 6 \times 17^2 = 1734$ cm^2 (1); Edge for volume 5832: $\sqrt[3]{5832} = 18$ cm (1).	1+1+1
	OR: Volume $= lbh = 5 \times 12 \times 13 = 780$ cm^3 (1); TSA $= 2(lb + bh + hl) = 2(60 + 156 + 65) = 562$ cm^2 (1); LSA $= 2h(l + b) = 442$ cm^2 (1).	1+1+1
18	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 3.5 \times 7 = 154$ cm^2 (1); TSA $= 2\pi r(r + h) = 231$ cm^2 (1); Volume $= \pi r^2 h = 269.5$ cm^3 (1).	1+1+1
	OR: Volume $= 16^3 = 4096$ cm^3 (1); TSA $= 6 \times 16^2 = 1536$ cm^2 (1); Edge for volume 4913: $\sqrt[3]{4913} = 17$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 21 \times 5 = 660$ cm^2 (1); TSA $= 2\pi r(r + h) = 3432$ cm^2 (2); Volume $= \pi r^2 h = 6930$ cm^3 (2).	1+2+2
	OR: Volume of cuboid $= 15 \times 30 \times 25 = 11250$ cm^3 (1); Volume of one cube $= 5^3 = 125$ cm^3 (1); Number of cubes $= \frac{11250}{125} = 90$ (1); Leftover volume $= 0$ (the cuboid divides exactly) (2).	1+1+1+2
20	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 7 \times 16 = 704$ cm^2 (1); TSA $= 2\pi r(r + h) = 1012$ cm^2 (2); Volume $= \pi r^2 h = 2464$ cm^3 (2).	1+2+2
	OR: Volume of cuboid $= 20 \times 8 \times 12 = 1920$ cm^3 (1); Volume of one cube $= 4^3 = 64$ cm^3 (1); Number of cubes $= \frac{1920}{64} = 30$ (1); Leftover volume $= 0$ (the cuboid divides exactly) (2).	1+1+1+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in πrl (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.

- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 294 cm^2	$6a^2$.
2	**(a)** 198 cm^2	$2h(l+b)$.
3	**(d)** $4\pi r^2$	Standard formula.
4	**(d)** 5 cm	$\sqrt[3]{125}=5$.
5	**(c)** 231 cm^3	$\pi r^2 h$ with $r=3.5$, $h=6$.
6	**(a)** 154 cm^2	$4\pi r^2$ with $r=3.5$.
7	**(d)** 70 cm^3	$lbh=2\times5\times7$.
8	**(a)** 729 cm^3	9^3.
9	**(a)** 384 cm^2	$6a^2$.
10	**(c)** 180 cm^2	$2h(l+b)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume $=7^3=343$ cm^3; TSA $=6\times7^2=294$ cm^2 (1); Edge for volume 512: $\sqrt[3]{512}=8$ cm (1).	1+1
12	Volume $=5\times9\times2=90$ m^3; Capacity $=90\times1000$ litres (1); $=90000$ litres (1).	1+1
13	Volume $=lbh=9\times2\times8=144$ cm^3; TSA $=2(lb+bh+hl)=2(18+16+72)=212$ cm^2 (1); LSA $=2h(l+b)=176$ cm^2 (1).	1+1
14	Volume $=9^3=729$ cm^3; TSA $=6\times9^2=486$ cm^2 (1); Edge for volume 1000: $\sqrt[3]{1000}=10$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume $=lbh=10\times8\times11=880$ cm^3 (1); TSA $=2(lb+bh+hl)=2(80+88+110)=556$ cm^2 (1); LSA $=2h(l+b)=396$ cm^2 (1).	1+1+1
16	CSA $=2\pi rh=2\times\frac{22}{7}\times14\times5=440$ cm^2 (1); TSA $=2\pi r(r+h)=1672$ cm^2 (1); Volume $=\pi r^2h=3080$ cm^3 (1).	1+1+1
17	Volume $=14^3=2744$ cm^3 (1); TSA $=6\times14^2=1176$ cm^2 (1); Edge for volume 3375: $\sqrt[3]{3375}=15$ cm (1).	1+1+1
	OR: Volume $=lbh=5\times6\times9=270$ cm^3 (1); TSA $=2(lb+bh+hl)=2(30+54+45)=258$ cm^2 (1); LSA $=2h(l+b)=198$ cm^2 (1).	1+1+1
18	CSA $=2\pi rh=2\times\frac{22}{7}\times14\times11=968$ cm^2 (1); TSA $=2\pi r(r+h)=2200$ cm^2 (1); Volume $=\pi r^2h=6776$ cm^3 (1).	1+1+1
	OR: Volume $=18^3=5832$ cm^3 (1); TSA $=6\times18^2=1944$ cm^2 (1); Edge for volume 6859: $\sqrt[3]{6859}=19$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	CSA $=2\pi rh=2\times\frac{22}{7}\times14\times12=1056$ cm^2 (1); TSA $=2\pi r(r+h)=2288$ cm^2 (2); Volume $=\pi r^2h=7392$ cm^3 (2).	1+2+2
	OR: Volume of cuboid $=4\times6\times8=192$ cm^3 (1); Volume of one cube $=2^3=8$ cm^3 (1); Number of cubes $=\frac{192}{8}=24$ (1); Leftover volume $=0$ (the cuboid divides exactly) (2).	1+1+1+2
20	CSA $=2\pi rh=2\times\frac{22}{7}\times10.5\times8=528$ cm^2 (1); TSA $=2\pi r(r+h)=1221$ cm^2 (2); Volume $=\pi r^2h=2772$ cm^3 (2).	1+2+2
	OR: Volume of cuboid $=20\times24\times16=7680$ cm^3 (1); Volume of one cube $=4^3=64$ cm^3 (1); Number of cubes $=\frac{7680}{64}=120$ (1); Leftover volume $=0$ (the cuboid divides exactly) (2).	1+1+1+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in πrl (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi rh+\pi r^2$; there are two circular ends, so it is $2\pi rh+2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div1000$, and m^3 to litres needs $\times1000$.

- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $2\pi r^2$	Half of a sphere's surface.
2	**(b)** 8 cm	$\sqrt[3]{512} = 8$.
3	**(a)** 924 cm^3	$\pi r^2 h$ with $r = 7$, $h = 6$.
4	**(c)** 2464 cm^2	$4\pi r^2$ with $r = 14$.
5	**(a)** 360 cm^3	$lbh = 5 \times 8 \times 9$.
6	**(b)** 8 cm^3	2^3.
7	**(b)** 216 cm^2	$6a^2$.
8	**(b)** 70 cm^2	$2h(l + b)$.
9	**(b)** $l^2 = r^2 + h^2$	Right triangle.
10	**(a)** 7 cm	$\sqrt[3]{343} = 7$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume $= 8 \times 5 \times 9 = 360$ m^3; Capacity $= 360 \times 1000$ litres (1); $= 360000$ litres (1).	1+1
12	Volume $= lbh = 8 \times 9 \times 6 = 432$ cm^3; TSA $= 2(lb + bh + hl) = 2(72 + 54 + 48) = 348$ cm^2 (1); LSA $= 2h(l + b) = 204$ cm^2 (1).	1+1
13	Volume $= 8^3 = 512$ cm^3; TSA $= 6 \times 8^2 = 384$ cm^2 (1); Edge for volume 729: $\sqrt[3]{729} = 9$ cm (1).	1+1
14	Volume $= 9 \times 6 \times 7 = 378$ m^3; Capacity $= 378 \times 1000$ litres (1); $= 378000$ litres (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume $= lbh = 7 \times 14 \times 9 = 882$ cm^3 (1); TSA $= 2(lb + bh + hl) = 2(98 + 126 + 63) = 574$ cm^2 (1); LSA $= 2h(l + b) = 378$ cm^2 (1).	1+1+1
16	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 3.5 \times 4 = 88$ cm^2 (1); TSA $= 2\pi r(r + h) = 165$ cm^2 (1); Volume $= \pi r^2 h = 154$ cm^3 (1).	1+1+1
17	Volume $= 20^3 = 8000$ cm^3 (1); TSA $= 6 \times 20^2 = 2400$ cm^2 (1); Edge for volume 9261: $\sqrt[3]{9261} = 21$ cm (1).	1+1+1
	OR: Volume $= lbh = 8 \times 12 \times 11 = 1056$ cm^3 (1); TSA $= 2(lb + bh + hl) = 2(96 + 132 + 88) = 632$ cm^2 (1); LSA $= 2h(l + b) = 440$ cm^2 (1).	1+1+1
18	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 14 \times 10 = 880$ cm^2 (1); TSA $= 2\pi r(r + h) = 2112$ cm^2 (1); Volume $= \pi r^2 h = 6160$ cm^3 (1).	1+1+1
	OR: Volume $= 13^3 = 2197$ cm^3 (1); TSA $= 6 \times 13^2 = 1014$ cm^2 (1); Edge for volume 2744: $\sqrt[3]{2744} = 14$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 21 \times 11 = 1452$ cm^2 (1); TSA $= 2\pi r(r + h) = 4224$ cm^2 (2); Volume $= \pi r^2 h = 15246$ cm^3 (2).	1+2+2
	OR: Volume of cuboid $= 9 \times 12 \times 6 = 648$ cm^3 (1); Volume of one cube $= 3^3 = 27$ cm^3 (1); Number of cubes $= \frac{648}{27} = 24$ (1); Leftover volume $= 0$ (the cuboid divides exactly) (2).	1+1+1+2
20	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 10.5 \times 18 = 1188$ cm^2 (1); TSA $= 2\pi r(r + h) = 1881$ cm^2 (2); Volume $= \pi r^2 h = 6237$ cm^3 (2).	1+2+2
	OR: Volume of cuboid $= 16 \times 24 \times 8 = 3072$ cm^3 (1); Volume of one cube $= 4^3 = 64$ cm^3 (1); Number of cubes $= \frac{3072}{64} = 48$ (1); Leftover volume $= 0$ (the cuboid divides exactly) (2).	1+1+1+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in πrl (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.

- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
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